**Table A1.1** Systematic developments in physics over centuries (*Cont*)

| | |
|---|---|
| ▪ Copernicun Revolution<br>▪ Heliocentric model (Not hypothesis. It provides simplest explanation than Ptolemy model for motion of stars and Planets )<br>▪ Accurate astronomical datas<br>▪ Laws of Planetary motion | 15th Century AD<br>AD 1543 (Copernicus)<br><br><br><br>Tycho Brahe<br>Kepler |
| ▪ Law of inertia<br>▪ Telescope observation(Founder of modern observational astronomy)<br>▪ Calculation of time period of the Moons of the Jupiter<br>▪ Earth is not flat<br>▪ All object fall to Earth at the same rate (Disproved Aristotle's argument)<br>▪ Law of inertia(Force need not required to maintain the motion- disproved Aristotle's argument)<br>▪ Pendulum, inclined plane experiments<br>▪ Study of projectile motion<br>▪ Introduction of 'controlled experiments' | Gallieo Galilei (1564- 1642) |
| ▪ Introduction of Cartesian coordinate system<br>▪ Idea of analytical geometry | Rane Descarte (1596-1650) |
| ▪ 17th and 18thcentury Development | |
| ▪ Laws of motion<br>▪ Quantitative idea of motion<br>▪ Law of gravitation<br>▪ Development of calculus independent of Leibnitz<br>▪ Founder of modern optics (reflection, dispersion, prism)<br>▪ Light consists of minute particles 'corpuscules'<br>▪ Derivation of Kepler laws<br>▪ Greatest book 'The principia mathematica'(1687)<br>▪ Wave theory of light | Isaac Newton (1642-1727)<br><br><br><br><br><br><br><br><br><br>Christian Huygens; Father of modern optics |



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